

Chain Rule

Section 2.4

9-5-25

"People who say it can not be done should not interrupt those who are doing it"

George Bernard Shaw

2 Types of functions

Plain Functions

$$y = 5x^7$$

$$f(x) = \sin x$$

$$y = \cos x$$

Don't need chain rule

Composite Functions

$$y = (3x^2 + 5)^7$$

$$f(x) = \sin(4x)$$

$$y = \cos^3 x$$

Needs CHAIN RULE

CHAIN RULE

(yes, I know I already said it.)

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Ex 1

$$f(x) = (4x^3 + 3x - 2)^5$$

$$f'(x) = ?$$

$$f'(x) = 5(4x^3 + 3x - 2)^4 \cdot (12x^2 + 3)$$

Ex 2

$$f(x) = (x^4 + 2)^{2/3}$$

$$f'(x) = ?$$

$$f'(x) = \frac{2}{3}(x^4 + 2)^{-1/3} \cdot (4x^3)$$

$$f'(x) = \frac{2}{3(x^4 + 2)^{1/3}} \cdot 4x^3$$

$$f'(x) = \frac{8x^3}{3\sqrt[3]{x^4 + 2}}$$

Ex 3

$$y = \frac{-1}{(2x-3)^2}$$

$$y' = ?$$

$$y' = 14(2x-3)^{-3} \cdot 2 = \frac{28}{(2x-3)^3}$$

EX 4 CHALLENGE

$$h(x) = (f(5x))^4$$

$$h'(x) = ?$$

$$h'(x) = 4(f(5x))^3 \cdot f'(5x) \cdot 5$$

TRIG CHAIN

(Examples)

Ex 1

$$y = \cos(3x)$$

$$y' = ?$$

$$y' = -\sin(3x) \cdot 3$$

$$-3\sin(3x)$$

Ex 2

$$f(x) = \sin(5x^2)$$

$$f'(x) = ?$$

$$f'(x) = \cos(5x^2) \cdot 10x$$

$$10x \cos(5x^2)$$

Ex 3

$$y = \csc(7x^5)$$

$$y' = ?$$

$$y' = -\csc(7x^5) \cot(7x^5) \cdot 35x^4$$

$$-35x^4 (\csc(7x^5) \cot(7x^5))$$

Ex 4

$$y = \cos^3 x = (\cos x)^3$$

$$y' = ?$$

$$y' = 3(\cos x)^2 \cdot -\sin x$$

$$-3(\cos x)^2 \cdot \sin x$$

Ex 5

$$y = \cos^4(5x)$$

$$y' = ?$$

$$y = (\cos(5x))^4$$

$$y' = 4(\cos(5x))^3 \cdot -\sin(5x) \cdot 5$$

$$-20(\cos^3(5x) \sin(5x))$$

Ex 6

$$f(x) = \sec^2(4x^3 + 5)$$

$$f'(x) = ?$$

$$f(x) = (\sec(4x^3 + 5))^2$$

$$f'(x) = 2(\sec(4x^3 + 5)) \cdot \sec(4x^3 + 5) \tan(4x^3 + 5) \cdot 12x^2$$

$$24x^2 (\sec^2(4x^3 + 5) \tan(4x^3 + 5))$$

And thus ends the

the chain rule

notes...